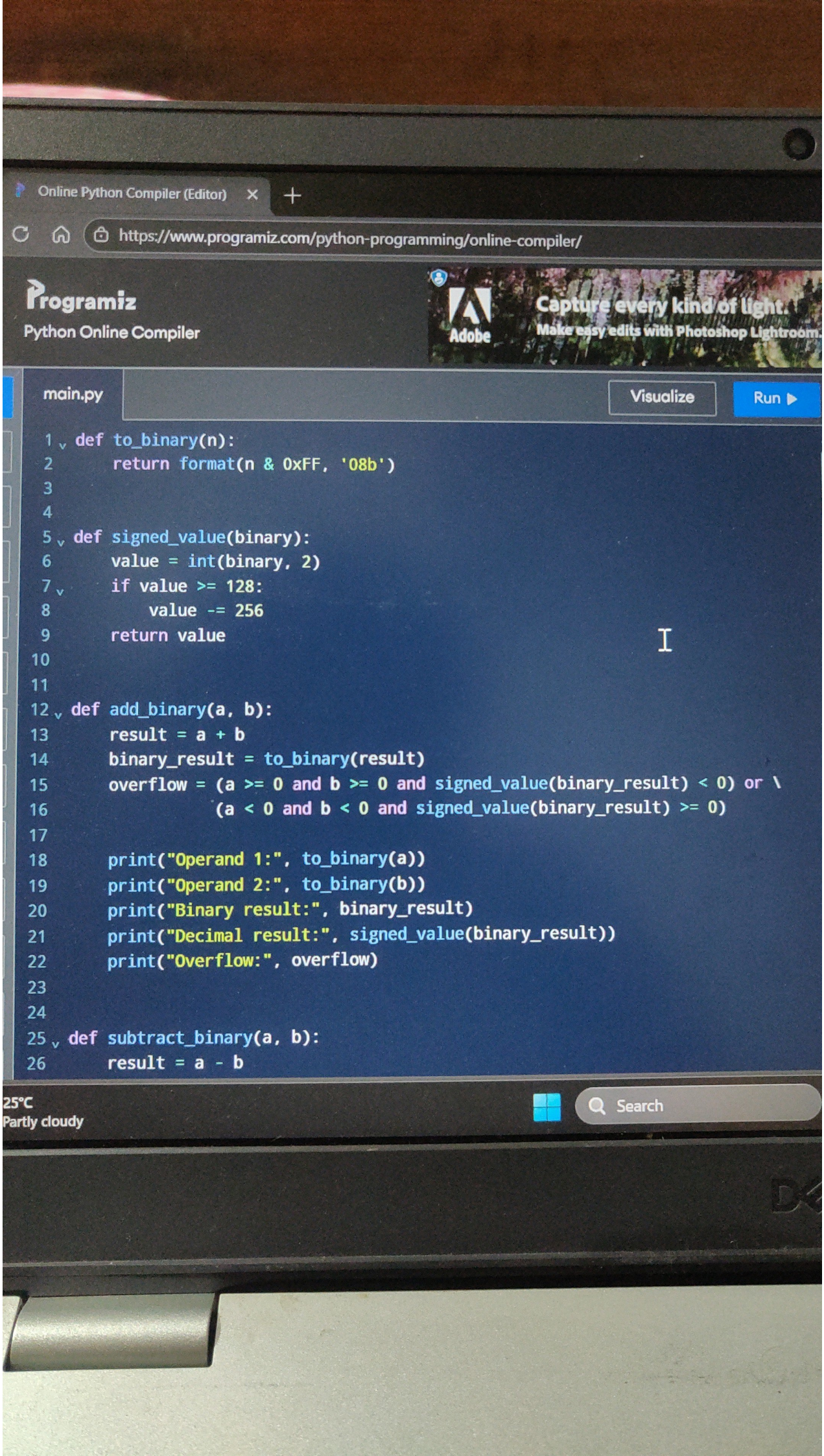
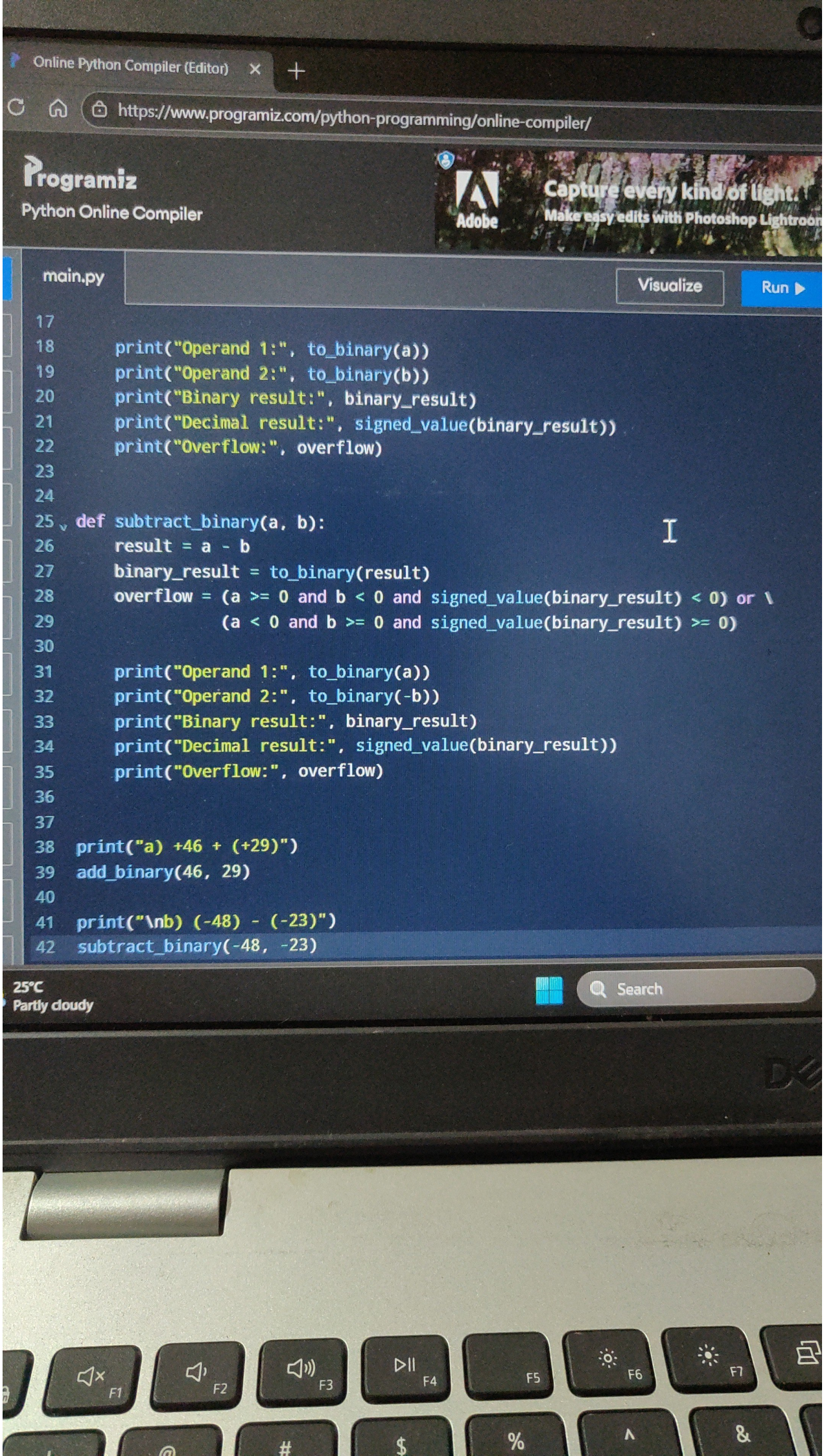

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def to_binary(n):
return format(n & 0xFF, '08b')
def signed_value(binary):
value = int(binary, 2)
if value >= 128:
value -= 256
return value
def add_binary(a, b):
result = a + b
binary_result = to_binary(result)
overflow = (a >= 0 and b >= 0 and signed_value(binary_result) < 0) or \
(a < 0 and b < 0 and signed_value(binary_result) >= 0)
print("Operand 1:", to_binary(a))
print("Operand 2:", to_binary(b))
print("Binary result:", binary_result)
print("Decimal result:", signed_value(binary_result))
print("Overflow:", overflow)
def subtract_binary(a, b):
result = a - b
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main.py
Visualize
Run
print("Operand 1:", to_binary(a))
print("Operand 2:", to_binary(b))
print("Binary result:", binary_result)
print("Decimal result:", signed_value(binary_result))
print("Overflow:", overflow)
def subtract_binary(a, b):
result = a - b
binary_result = to_binary(result)
overflow = (a >= 0 and b < 0 and signed_value(binary_result) < 0) or \
(a < 0 and b >= 0 and signed_value(binary_result) >= 0)
print("Operand 1:", to_binary(a))
print("Operand 2:", to_binary(-b))
print("Binary result:", binary_result)
print("Decimal result:", signed_value(binary_result))
print("Overflow:", overflow)
print("a) +46 + (+29)")
add_binary(46, 29)
print("\nb) (-48) - (-23)")
subtract_binary(-48, -23)
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Output

```
a) +46 + (+29)
Operand 1: 00101110
Operand 2: 00011101
Binary result: 01001011
Decimal result: 75
Overflow: False

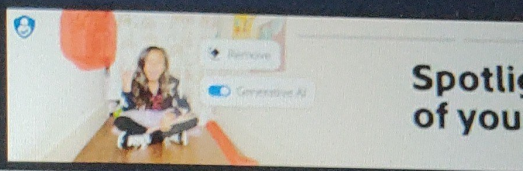
b) (-48) - (-23)
Operand 1: 11010000
Operand 2: 00010111
Binary result: 11100111
Decimal result: -25
Overflow: False
```

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```python
def rotate_right(binary, positions):
    print("Original number:", binary)
    current = binary

    for i in range(positions):
        last_bit = current[-1]
        current = last_bit + current[:-1]
        print("Step", i + 1, ":", current)

    print("Final result:", current)


def rotate_right_carry(binary, positions, carry):
    print("Original number:", binary)
    print("Initial Carry Flag:", carry)
    current = binary

    for i in range(positions):
        new_carry = int(current[-1])
        current = str(carry) + current[:-1]
        carry = new_carry

        print("Step", i + 1, ":", current
```

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Ru

```
def rotate_right_carry(binary, positions, carry):
    print("Original number:", binary)
    print("Initial Carry Flag:", carry)
    current = binary

    for i in range(positions):
        new_carry = int(current[-1])
        current = str(carry) + current[:-1]
        carry = new_carry

        print("Step", i + 1, ":", current,
              "Carry Flag:", carry)

    print("Final result:", current)
    print("Final Carry Flag:", carry)


print("a) ROR")
rotate_right("11010101", 3)

print("\nb) RCR")
rotate_right_carry("11010101", 3, 0)
```

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Output

```
a) ROR
Original number: 11010101
Step 1 : 11101010
Step 2 : 01110101
Step 3 : 10111010
Final result: 10111010

b) RCR
Original number: 11010101
Initial Carry Flag: 0
Step 1 : 01101010 Carry Flag: 1
Step 2 : 10110101 Carry Flag: 0
Step 3 : 01011010 Carry Flag: 1
Final result: 01011010
Final Carry Flag: 1
```

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